Submission Summary

Conference Name
IEEE International Conference on Power Electronics, Drives and Energy Systems 2026 (PEDES 2026)

Paper ID
1363

Paper Title
Constrained Optimization Based Design and Control of Efficient EV Drives

Abstract
Interior permanent magnet (IPM) motors are extensively employed in electric vehicle (EV) traction systems because of their high efficiency and power density. During EV operation, the motor covers a wide range of speeds and torque demands, including maximum torque per ampere (MTPA), flux-weakening (FW), and maximum torque per voltage (MTPV) regions. Accurate d-axis current reference generation is essential for achieving optimal performance across these operating regions. Conventional analytical approaches generally use predefined current trajectories, which can result in inaccurate negative d-axis current in voltage-limited regions, increasing stator current and conduction losses. Although online optimization can improve the operating point, its computational burden limits real-time implementation, whereas complete drive-cycle lookup tables require considerable memory. This paper proposes a selective constrained optimization approach for d-axis current reference generation. An analytical solution is retained for the MTPA region, while offline optimized lookup tables are generated only for the FW and MTPV regions. A dynamic constrained demagnetization algorithm reduces the optimization to a single decision variable. By restricting optimization to voltage-limited regions, the proposed approach significantly reduces memory and computational requirements while enabling real-time implementation. Simulation and experimental results confirm the effectiveness of the proposed control architecture.

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Primary Subject Area

Track 03: Electrical Machines and Industrial Drives

Conflicts of Interest

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Submission Files

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Submission Questions Response

1. Agreement to the guidelines

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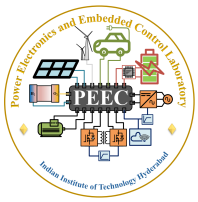
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3. Author list

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Regards,
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PhD Scholar,
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To: Akshay Chabukswar <ee21resch01003@iith.ac.in>

Dear Akshay Chabukswar,

Congratulations!!

We are pleased to inform you that your paper ID #1363 titled "Constrained Optimization Based Design and Control of Efficient EV Drives" has been accepted in IEEE International Conference on Power Electronics, Drives and Energy Systems 2026 (PEDES 2026). The conference will be held from 16th to 19th December 2026 at IIT Guwahati, Assam, India.

The reviews can be checked at the author console on the Microsoft CMT portal. Please incorporate suggestions from the reviewers in the final camera-ready submission.

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Thank you very much for your contribution. We look forward to seeing you at the conference.

Regards,
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